# Python Libraries Used in Machine Learning

## Overview

This page lists the most commonly used Python libraries in Machine Learning, along with their primary use cases and where they are typically applied.

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# Core Machine Learning Libraries

| Library | Primary Use Case | Common Applications |
|---|---|---|
| **NumPy** | Numerical computing, arrays, matrix operations | Data preprocessing, mathematical operations, linear algebra |
| **Pandas** | Data loading, cleaning, manipulation | CSV/Excel processing, EDA, feature engineering |
| **Matplotlib** | Basic data visualization | Line plots, bar charts, histograms, scatter plots |
| **Seaborn** | Statistical visualization | Correlation heatmaps, pair plots, distribution plots |
| **Plotly** | Interactive visualizations | Dashboards, interactive graphs, presentations |
| **Scikit-learn (sklearn)** | Traditional Machine Learning | Classification, Regression, Clustering, Preprocessing, Model Evaluation |
| **SciPy** | Scientific computing | Optimization, statistics, signal processing |
| **TensorFlow** | Deep Learning framework | Neural Networks, CNNs, RNNs, Transformers |

| Library | Primary Use Case | Common Applications |
|---|---|---|
| **PyTorch** | Deep Learning framework | Research, LLMs, Computer Vision, NLP |
| **XGBoost** | Gradient Boosting | Tabular datasets, Kaggle competitions |
| **LightGBM** | Fast Gradient Boosting | Large datasets, ranking problems |
| **CatBoost** | Gradient Boosting with categorical support | Datasets containing categorical features |
| **OpenCV (cv2)** | Image and video processing | Face detection, object detection, Computer Vision |
| **Pillow (PIL)** | Image processing | Image resizing, cropping, augmentation |
| **NLTK** | Natural Language Processing | Tokenization, stemming, text preprocessing |
| **spaCy** | Industrial NLP pipeline | Named Entity Recognition, POS tagging |
| **Transformers (Hugging Face)** | Pretrained AI models | BERT, GPT, T5, Llama, Text Generation |
| **Datasets (Hugging Face)** | Dataset management | NLP and Machine Learning datasets |
| **TensorFlow Datasets (TFDS)** | Ready-to-use datasets | TensorFlow projects |
| **Joblib** | Model serialization | Saving and loading ML models |
| **Pickle** | Object serialization | Saving Python objects and models |
| **Statsmodels** | Statistical analysis | Hypothesis testing, regression, time-series analysis |
| **Imbalanced-learn (imblearn)** | Imbalanced dataset handling | SMOTE, over-sampling, under-sampling |
| **SHAP** | Model explainability | Feature importance, prediction interpretation |
| **LIME** | Local model explanation | Explaining individual predictions |

| Library | Primary Use Case | Common Applications |
|---|---|---|
| **Optuna** | Hyperparameter optimization | Automatic model tuning |

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# Typical Machine Learning Workflow

1. Load Dataset (Pandas)
2. Data Cleaning (Pandas + NumPy)
3. Exploratory Data Analysis (Matplotlib + Seaborn + Plotly)
4. Feature Engineering (Pandas + NumPy)
5. Train-Test Split (Scikit-learn)
6. Model Training (Scikit-learn, XGBoost, LightGBM, CatBoost, TensorFlow, PyTorch)
7. Model Evaluation (Scikit-learn)
8. Model Explainability (SHAP / LIME)
9. Model Saving (Joblib / Pickle)

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# Libraries by Domain

## 📊 Tabular Data

- Pandas
- NumPy
- Scikit-learn
- XGBoost
- LightGBM
- CatBoost

## 🖼️ Computer Vision

- OpenCV
- Pillow
- TensorFlow
- PyTorch

## 💬 Natural Language Processing (NLP)

- NLTK
- spaCy
- Hugging Face Transformers
- Hugging Face Datasets

## 📈 Data Visualization

- Matplotlib
- Seaborn
- Plotly

## 🤖 Deep Learning

- TensorFlow
- PyTorch

## ⚡ Model Optimization

- Optuna
- SciPy

## 💾 Model Deployment

- Joblib
- Pickle

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# Essential Libraries for Beginners

If you're starting Machine Learning, master these libraries first:

1. NumPy
2. Pandas
3. Matplotlib
4. Seaborn
5. Scikit-learn
6. XGBoost
7. TensorFlow **or** PyTorch
8. Hugging Face Transformers (for NLP/LLMs)